AEM8202: Fluid Mechanics II **Spring 2026**

<u>Take-home final exam</u>

Due Wed., May 13th by 11:59pm

Please work independently on the following multi-part question. Submit completed exams as a single PDF file to Canvas by 11:59pm on Wed., May 13th.

Question 1: We have seen in lecture that the Joukowski transform

$$z = \zeta + \frac{1}{\zeta},$$

maps flow around a cylinder of radius $a > 1$ in the ζ-plane to flow around an ellipse in the z-plane. Similarly, flow around a "Joukowski" airfoil may be constructed by using the transform to map flow around an off-center cylinder in the ζ-plane. For a symmetric airfoil, the center m of the off-center cylinder lies on the negative ζ-axis, and the radius $a = 1 + m$ so that the cylinder passes through the point $\zeta = 1$, forming cusp-like trailing edge in the z-plane.

a) Find a value for m such that the airfoil's maximum thickness t_{max} (in the y-direction) is 4% and then of its chord length ℓ. Repeat to find m for a Joukowski airfoil with a thickness of 8%. It is possible to do this numerically by discretizing appropriate circles in the ζ-plane and passing them through the transform to evaluate the thickness in the z-plane, and then adjusting m until you get the desired results. There are also expressions in Ch. 14 of Kundu that can help. Note that the maximum thickness does not necessarily occur at $x = 0$.

b) Find an expression for the complex potential function $F(\zeta(z))$ for flow around a Joukowski airfoil. Using your favorite software, plot streamlines around the airfoil with 8% thickness. Note: To avoid branch cuts, define $\zeta(z) = \frac{1}{2}\left(z + \sqrt{z-2}\sqrt{z+2}\right)$, evaluate ζ first, and then evaluate $F(\zeta)$.

c) For both the 4% and 8% airfoils, plot the tangential velocity as a function of arclength along the top surface of each airfoil, starting at the stagnation point and ending at the trailing edge. Note: Arclength can be found numerically by integrating $|dz|$. Assume $U_\infty = 1$.

d) Now consider a non-zero angle-of-attack by rotating the flow in the ζ-plane by a counterclockwise angle α. Find an expression for $F(\zeta)$ that will satisfies the Kutta condition in the z-plane. Using your favorite software, plot streamlines around the 8% thickness airfoil at $\alpha = 4°$ angle-of-attack.

e) Assuming $\rho = 1$, find the magnitude of the lift force (perpendicular to the freestream flow) acting on the airfoil for the case in part d using Blasius' Theorem. Verify your answer by numerically integrating $i\frac{\rho}{2}\oint W^2 dz$. The chain rule may help you here.

f) Assuming a kinematic viscosity of $\nu = 0.001$ and that the flow remains laminar over the entire airfoil, use your results in part (c) and Thwaites' method to find (numerically) the momentum thickness θ of the boundary layer that forms near each airfoil. Plot $\theta(s)$ and $\lambda(s) = \frac{\theta^2}{\nu}\frac{dU_\infty}{ds}$ as functions of the arclength s along the top surface of the airfoil starting at the stagnation point. Will the boundary layer remain attached to the airfoil in all three cases? Why or why not?

g) Now consider slowly increasing the angle-of-attack for the airfoils with 4% and 8% thicknesses. For each airfoil, make a graph containing plots of $\lambda(s)$ for $\alpha = 0°,\ 2°$, and $4°$. At what position does Thwaites' method predict the boundary layer will first separate from the airfoil in each case? Using your results, discuss the benefits and drawbacks of thickness in airfoil design. How do you think the shape of the Joukowski airfoil could be improved?